

EDA & Data Cleaning: Cars93 Dataset

Project Objective

This project performs exploratory data analysis (EDA) and data cleaning on the **Cars93** dataset — a collection of 93 car models from the 1993 model year with intentionally-introduced missing values. The goal is to inspect the raw data, identify and handle data quality issues (missing values), and produce a clean, analysis-ready dataset along with visualizations and a written summary of findings.

Dataset Source

- **Dataset:** Cars93 (with missing values)
- **Source:** [selva86/datasets](#) — Cars93_miss.csv
- **Rows / Columns:** 93 rows × 27 columns
- **Description:** Each row represents a 1993 car model, with attributes including manufacturer, price, engine specs, fuel efficiency (MPG), dimensions, weight, and passenger capacity.

Repository Structure

```
eda-cleaning-project/  
├── README.md  
├── data/  
│   ├── raw_dataset.csv          #  
Original dataset with missing values  
│   └── cleaned_dataset.csv      #  
Cleaned dataset (exported from the  
notebook)  
├── notebooks/  
│   └── eda_and_cleaning.ipynb   # Full  
EDA + cleaning workflow with code and  
outputs  
├── docs/  
│   └── summary_report.pdf       #  
Dataset overview, cleaning steps, insights,  
challenges  
└── .gitignore
```

How to Run the Notebook

1. Clone this repository:

```
git clone <your-repo-url>  
cd eda-cleaning-project
```

2. Install the required packages:

```
pip install pandas numpy matplotlib
```

```
seaborn jupyter
```

3. Launch Jupyter and open the notebook:

```
jupyter notebook  
notebooks/eda_and_cleaning.ipynb
```

4. Run all cells in order (Cell → Run All). This will:

- Load `data/raw_dataset.csv`
- Inspect shape, data types, missing values, and duplicates
- Clean the data (median imputation for numeric columns, mode imputation for categorical columns)
- Save the cleaned dataset to `data/cleaned_dataset.csv`
- Generate visualizations (missing value chart, price distribution, boxplot, correlation heatmap)

Summary

See [docs/summary_report.pdf](#) for a full written summary of the dataset, cleaning steps, key insights, and challenges encountered.